

## Project Executable Files

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|----------------------|-------------------|
| <b>Date</b>          | 30 September 2026 |
| <b>Team ID</b>       | PER206AIFITTRACK  |
| <b>Project Name</b>  | AI-FitTrack-API   |
| <b>Maximum Marks</b> | 3 Marks           |

### Project Executable Files:

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

#### Step 1: Submission Checklist

| S.No | Artifact / Deliverable Item                                           | Submitted ? | Location / File Details                                                    |
|------|-----------------------------------------------------------------------|-------------|----------------------------------------------------------------------------|
| 1    | Complete Backend Source Code (server.js, controllers, models, routes) | Yes         | AI-FitTrack-API/ root directory                                            |
| 2    | Project Setup Guide & Execution Instructions (README.md)              | Yes         | AI-FitTrack-API/README.md                                                  |
| 3    | Package & Dependency Manifest (package.json, package-lock.json)       | Yes         | AI-FitTrack-API/package.json (express, mongoose, jwt, @google/genai)       |
| 4    | Database Schema Models (User.js, Workout.js)                          | Yes         | AI-FitTrack-API/models/                                                    |
| 5    | Environment Configuration Template (.env.example)                     | Yes         | AI-FitTrack-API/.env.example (PORT, MONGO_URI, JWT_SECRET, GEMINI_API_KEY) |
| 6    | Automated Test Suite Script (test-suite.js)                           | Yes         | AI-FitTrack-API/test-suite.js (59 passing test assertions)                 |
| 7    | Full Project Documentation Report (DOCX / PDF)                        | Yes         | FitTrack_AI_Project_Report.docx & Sample Project Documentation.pdf         |
| 8    | Thunder Client API Testing Verification Screenshots PDF               | Yes         | FitTrack_API_Testing_Screenshots.pdf (10 verified endpoints)               |

#### Step 2: Project File & Folder Architecture

| File / Directory                 | Purpose in Application                                                             |
|----------------------------------|------------------------------------------------------------------------------------|
| server.js                        | Application entry point; initializes Express server on Port 5000 and mounts routes |
| package.json                     | Project dependencies, start script (node server.js), and test runner               |
| .env                             | Environment variables for PORT, MONGO_URI, JWT_SECRET, and GEMINI_API_KEY          |
| config/db.js                     | Asynchronous MongoDB database connection handler using Mongoose ODM                |
| models/User.js                   | Mongoose schema for user accounts, credentials, and fitness profiles               |
| models/Workout.js                | Mongoose schema for workout session logs with duration and calories                |
| middleware/authMiddleware.js     | JWT token extraction, signature verification, and route protection                 |
| controllers/authController.js    | Business logic for User Register, User Login, and Profile fetch                    |
| controllers/workoutController.js | CRUD business logic and search/filter query processors for workouts                |
| controllers/aiController.js      | Google Gemini 1.5 Flash prompt synthesis and generative fitness routines           |
| routes/*.js                      | REST route mappings for auth, workout, and AI services                             |

### Step 3: Local Execution & Verification Instructions

| Step                   | Command Line Instruction                  | Expected System Behavior & Output                                         |
|------------------------|-------------------------------------------|---------------------------------------------------------------------------|
| 1. Clone / Navigate    | cd AI-FitTrack-API                        | Enters root application project folder                                    |
| 2. Install Packages    | npm install                               | Installs express, mongoose, jsonwebtoken, bcryptjs, dotenv, @google/genai |
| 3. Configure .env      | cp .env.example .env (add GEMINI_API_KEY) | Configures Port 5000, MongoDB URI, JWT secret, and Google AI key          |
| 4. Start Server        | npm start                                 | Logs: MongoDB Connected: localhost:27017 & Server running on port 5000    |
| 5. Run Automated Tests | npm test (node test-suite.js)             | Executes 59 automated test assertions with 100% pass status               |
| 6. Thunder Client Test | Import Thunder Client Collection & Run    | Executes all 10 API requests with Status 200/201 responses                |